

## SUNDSVALL TIMRA, Sweden

WMO: 023660

Lat: 62.528N

Lon: 17.444E

Elev: 16

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-7.3 | (c)<br>-2.6 | (d)<br>-11.0                  | (e)<br>3.0 | (f)<br>-6.9 | (g)<br>-6.2 | (h)<br>4.0 | (i)<br>-2.3 | (j)<br>25.7           | (k)<br>32.2 | (l)<br>22.5 | (m)<br>29.7 | (n)<br>3.4            | (o)<br>310 | (p)<br>0.663 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.4                   | 76.7            | 63.5 | 73.3 | 61.9 | 70.1 | 60.2 | 65.8                | 72.8 | 63.8 | 70.1 | 62.0 | 67.8 | 9.5                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 63.2                            | 86.7 | 68.2 | 61.2 | 80.8 | 66.2 | 59.4 | 75.6 | 64.4 | 30.6          | 72.9 | 29.1 | 70.2 | 27.8 | 67.9 |                   |
|                                 |      |      |      |      |      |      |      |      |               |      |      |      |      |      |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |          | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|----------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |          | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max      | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 20.9          | (b) 18.0 | (c) 15.6 | (e) -13.0                  | (f) 82.0 | (g) 6.8            | (h) 4.2  | (i) -17.9                                          | (j) 85.0  | (k) -21.8  | (l) 87.5  | (m) -25.6  | (n) 89.9  | (o) -30.6  | (p) 92.9  |    |    |
| (4) 20.9          | (5) 18.0 | (6) 15.6 | (7) -13.3                  | (8) 67.6 | (9) 6.7            | (10) 2.2 | (11) -18.1                                         | (12) 69.2 | (13) -22.1 | (14) 70.5 | (15) -25.8 | (16) 71.7 | (17) -30.7 | (18) 73.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 39.6   | 20.2  | 20.9  | 28.5  | 37.1  | 46.3  | 55.2  | 61.1  | 58.8  | 50.9  | 40.1  | 30.7  | 23.7 |  |  |
|                                                                        | DBStd       | 15.99  | 10.67 | 11.02 | 8.14  | 5.31  | 5.95  | 4.89  | 4.65  | 4.79  | 5.24  | 5.75  | 7.24  | 9.92 |  |  |
|                                                                        | HDD50       | 4708   | 924   | 815   | 667   | 388   | 146   | 11    | 0     | 1     | 52    | 310   | 580   | 814  |  |  |
|                                                                        | HDD65       | 9319   | 1389  | 1235  | 1132  | 838   | 580   | 296   | 141   | 202   | 425   | 772   | 1030  | 1279 |  |  |
|                                                                        | CDD50       | 895    | 0     | 0     | 0     | 0     | 31    | 166   | 344   | 273   | 78    | 3     | 0     | 0    |  |  |
|                                                                        | CDD65       | 28     | 0     | 0     | 0     | 0     | 0     | 1     | 19    | 8     | 0     | 0     | 0     | 0    |  |  |
|                                                                        | CDH74       | 225    | 0     | 0     | 0     | 0     | 3     | 24    | 146   | 51    | 1     | 0     | 0     | 0    |  |  |
|                                                                        | CDH80       | 24     | 0     | 0     | 0     | 0     | 0     | 2     | 18    | 4     | 0     | 0     | 0     | 0    |  |  |
|                                                                        | WSAvg       | 6.3    | 5.7   | 5.6   | 6.4   | 6.5   | 7.2   | 7.6   | 6.8   | 6.5   | 6.2   | 6.0   | 5.8   | 5.6  |  |  |
|                                                                        | PrecAvg     | 26.3   | 2.1   | 1.3   | 1.3   | 1.5   | 1.5   | 2.2   | 3.1   | 3.3   | 2.6   | 2.7   | 2.4   | 2.2  |  |  |
|                                                                        | PrecMax     | 39.5   | 5.0   | 2.6   | 3.0   | 4.3   | 3.6   | 4.8   | 7.5   | 6.3   | 8.0   | 7.2   | 8.0   | 4.3  |  |  |
| (15) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | PrecMin     | 19.7   | 0.4   | 0.2   | 0.3   | 0.4   | 0.2   | 0.7   | 0.4   | 0.3   | 0.6   | 0.5   | 0.5   | 0.2  |  |  |
|                                                                        | PrecStd     | 4.7    | 1.1   | 0.6   | 0.7   | 1.0   | 0.8   | 1.1   | 1.8   | 1.6   | 1.7   | 1.7   | 1.7   | 1.0  |  |  |
|                                                                        | 0.4% DB     | 43.1   | 46.5  | 51.9  | 62.5  | 71.6  | 77.1  | 82.3  | 78.9  | 70.1  | 58.0  | 50.4  | 44.8  |      |  |  |
|                                                                        | 0.4% MCWB   | 38.7   | 40.0  | 43.0  | 49.2  | 56.7  | 61.5  | 65.1  | 64.1  | 59.6  | 51.5  | 45.9  | 40.8  |      |  |  |
|                                                                        | 2% DB       | 39.1   | 42.6  | 47.1  | 56.8  | 65.9  | 72.4  | 77.9  | 74.9  | 65.8  | 54.0  | 45.5  | 41.2  |      |  |  |
| (20) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% MCWB     | 35.6   | 37.9  | 39.8  | 46.2  | 53.5  | 58.4  | 64.1  | 63.5  | 57.9  | 49.6  | 43.6  | 38.0  |      |  |  |
|                                                                        | 5% DB       | 36.0   | 38.5  | 43.9  | 51.8  | 62.2  | 69.3  | 74.7  | 71.5  | 62.7  | 52.0  | 42.9  | 38.6  |      |  |  |
|                                                                        | 5% MCWB     | 34.2   | 34.6  | 37.7  | 43.0  | 51.5  | 57.2  | 63.1  | 62.1  | 56.2  | 48.7  | 41.7  | 36.8  |      |  |  |
|                                                                        | 10% DB      | 33.9   | 34.9  | 40.4  | 48.3  | 58.7  | 66.1  | 71.5  | 68.3  | 60.5  | 49.7  | 41.0  | 36.0  |      |  |  |
|                                                                        | 10% MCWB    | 32.9   | 32.9  | 35.3  | 41.0  | 49.0  | 55.6  | 61.4  | 60.2  | 55.1  | 46.7  | 39.6  | 34.9  |      |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | 39.4   | 40.6  | 43.9  | 49.9  | 58.5  | 63.6  | 68.7  | 68.3  | 62.0  | 53.9  | 46.9  | 41.3  |      |  |  |
|                                                                        | 0.4% MCDB   | 42.5   | 44.8  | 50.6  | 60.1  | 68.3  | 71.7  | 75.4  | 74.7  | 66.7  | 56.1  | 49.1  | 43.4  |      |  |  |
|                                                                        | 2% WB       | 36.3   | 38.0  | 40.4  | 46.7  | 55.1  | 60.7  | 66.6  | 65.6  | 59.7  | 51.6  | 43.9  | 38.8  |      |  |  |
|                                                                        | 2% MCDB     | 37.9   | 42.0  | 46.3  | 55.3  | 63.8  | 69.2  | 74.3  | 71.1  | 63.5  | 53.1  | 44.8  | 40.2  |      |  |  |
|                                                                        | 5% WB       | 34.5   | 35.0  | 38.0  | 43.9  | 52.3  | 58.9  | 64.7  | 63.6  | 57.7  | 49.3  | 41.9  | 36.7  |      |  |  |
| (33) Mean Daily Temperature Range                                      | 5% MCDB     | 35.7   | 37.3  | 43.0  | 50.7  | 60.3  | 65.9  | 72.0  | 68.6  | 61.2  | 51.3  | 42.6  | 38.1  |      |  |  |
|                                                                        | 10% WB      | 32.9   | 33.1  | 35.7  | 41.4  | 50.0  | 57.0  | 62.9  | 61.8  | 56.0  | 46.9  | 39.8  | 34.7  |      |  |  |
|                                                                        | 10% MCDB    | 33.8   | 34.5  | 39.6  | 46.7  | 56.7  | 63.6  | 69.3  | 66.8  | 59.3  | 48.9  | 40.7  | 35.8  |      |  |  |
|                                                                        | 5% DB       | 13.4   | 15.7  | 16.1  | 16.9  | 18.0  | 17.0  | 16.4  | 16.2  | 15.5  | 13.0  | 9.7   | 11.4  |      |  |  |
|                                                                        | 5% DB MCDBR | 15.1   | 18.2  | 20.0  | 25.3  | 24.8  | 23.0  | 21.5  | 20.3  | 18.8  | 17.0  | 12.1  | 13.9  |      |  |  |
| (40) Clear-Sky Solar Irradiance                                        | 5% DB MCWBR | 13.2   | 14.0  | 13.4  | 15.3  | 13.4  | 11.3  | 10.2  | 10.5  | 11.6  | 12.3  | 10.3  | 11.7  |      |  |  |
|                                                                        | 5% WB MCDBR | 14.1   | 16.1  | 19.0  | 23.7  | 22.9  | 19.6  | 18.6  | 17.3  | 16.5  | 14.5  | 10.9  | 12.4  |      |  |  |
|                                                                        | 5% WB MCWBR | 12.7   | 13.0  | 13.1  | 15.1  | 13.4  | 11.0  | 9.8   | 10.0  | 11.4  | 11.6  | 9.8   | 10.9  |      |  |  |
|                                                                        | taub        | 0.240  | 0.297 | 0.320 | 0.358 | 0.359 | 0.354 | 0.371 | 0.363 | 0.329 | 0.315 | 0.258 | 0.346 |      |  |  |
|                                                                        | taud        | 2.103  | 2.235 | 2.333 | 2.360 | 2.424 | 2.473 | 2.445 | 2.463 | 2.541 | 2.436 | 2.175 | 2.595 |      |  |  |
| (42) All-Sky Solar Radiation                                           | Ebn at Noon | 134    | 206   | 249   | 262   | 272   | 276   | 267   | 259   | 249   | 203   | 126   | 81    |      |  |  |
|                                                                        | Edh at Noon | 10     | 20    | 26    | 31    | 32    | 31    | 31    | 28    | 22    | 16    | 10    | 6     |      |  |  |
|                                                                        | RadAvg      | 63     | 267   | 734   | 1193  | 1520  | 1757  | 1604  | 1218  | 765   | 334   | 94    | 34    |      |  |  |
| (44) All-Sky Solar Radiation                                           | RadStd      | 8      | 43    | 82    | 102   | 157   | 111   | 173   | 102   | 94    | 60    | 19    | 3     |      |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +0.94 | N/A     | N/A    | N/A     | N/A   | N/A   | -268        | -299  | N/A   | N/A   |
| (47) Regional (35 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page